

<u>РБНФ</u>	<u>Код для перевірки РБНФ</u>
labeled_point = ident , ":";	labeled_point = ident >> tokenCOLON;
goto_label = "GOTO" , ident;	goto_label = tokenGOTO >> ident;
program_name = ident;	program_name = SAME_RULE(ident);
value_type = "INTEGER16";	value_type = SAME_RULE(tokenINTEGER16);
declaration_element = ident , ["[" , unsigned_value , "]"];	declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);
other_declaration_ident = "," , declaration_element;	other_declaration_ident = tokenCOMMA >> declaration_element;
declaration = value_type , declaration_element , {other_declaration_ident};	declaration = value_type >> declaration_element >> *other_declaration_ident;
index_action = "[" , expression , "]" ;	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;
unary_operator = "NOT";	unary_operator = SAME_RULE(tokenNOT);
unary_operation = unary_operator , expression;	unary_operation = unary_operator >> expression;
binary_operator = "AND" "OR" "==" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;
binary_action = binary_operator , expression;	binary_action = binary_operator >> expression;
left_expression = group_expression unary_operation ident , [index_action] value;	left_expression = group_expression unary_operation ident >> - index_action value;
expression = left_expression , {binary_action};	expression = left_expression >> *binary_action;
group_expression = "(" , expression , ")";	group_expression = tokenGROUPEXPRESSIONBEGIN >> expression >> tokenGROUPEXPRESSIONEND;
bind_right_to_left = ident , [index_action] , ":", expression;	bind_right_to_left = ident >> -index_action >> tokenRLBIND >> expression;
bind_left_to_right = expression , ":", ident , [index_action];	bind_left_to_right = expression >> tokenLRBIND >> ident >> - index_action;
if_expression = expression;	if_expression = SAME_RULE(expression);
body_for_true = block_statements_in_while_and_if_body;	body_for_true = SAME_RULE(block_statements_in_while_and_if_body);
false_cond_block_without_else = "ELSE" , cond_block;	false_cond_block_without_else = tokenELSE >> cond_block;
body_for_false = "ELSE" , block_statements_in_while_and_if_body;	body_for_false = tokenELSE >> block_statements_in_while_and_if_body;
cond_block = "IF" , if_expression , body_for_true ,	cond_block = tokenIF >> if_expression >> body_for_true >>

{false_cond_block_without_else} , [body_for_false];	*false_cond_block_without_else >> (-body_for_false);
cycle_begin_expression = expression;	cycle_begin_expression = SAME_RULE(expression);
cycle_end_expression = expression;	cycle_end_expression = SAME_RULE(expression);
cycle_counter = ident;	cycle_counter = SAME_RULE(ident);
cycle_counter_rl_init = cycle_counter , "!=" , cycle_begin_expression;	cycle_counter_rl_init = cycle_counter >> tokenRLBIND >> cycle_begin_expression;
cycle_counter_lr_init = cycle_begin_expression , "!=" , cycle_counter;	cycle_counter_lr_init = cycle_begin_expression >> tokenLRBIND >> cycle_counter;
cycle_counter_init = cycle_counter_rl_init cycle_counter_lr_init;	cycle_counter_init = cycle_counter_rl_init cycle_counter_lr_init;
cycle_counter_last_value = cycle_end_expression;	cycle_counter_last_value = SAME_RULE(cycle_end_expression);
cycle_body = "DO" , ({statement} block_statements);	cycle_body = tokenDO >> (statement block_statements);
forto_cycle = "FOR" , cycle_counter_init , ("TO" "DOWNT0") , cycle_counter_last_value , cycle_body;	forto_cycle = tokenFOR >> cycle_counter_init >> (tokenTO tokenDOWNT0) >> cycle_counter_last_value >> cycle_body;
	continue_while = SAME_RULE(tokenCONTINUE);
	break_while = SAME_RULE(tokenBREAK);
statement_in_while_and_if_body = statement "CONTINUE" "BREAK";	statement_in_while_and_if_body = statement continue_while break_while;
block_statements_in_while_and_if_body = "{" , {statement_in_while_and_if_body} , "}";	block_statements_in_while_and_if_body = tokenBEGINBLOCK >> *statement_in_while_and_if_body >> tokenENDBLOCK;
while_cycle_head_expression = expression;	while_cycle_head_expression = SAME_RULE(expression);
while_cycle = "WHILE" , while_cycle_head_expression , block_statements_in_while_and_if_body;	while_cycle = tokenWHILE >> while_cycle_head_expression >> block_statements_in_while_and_if_body;
repeat_until_cycle_cond = expression;	repeat_until_cycle_cond = SAME_RULE(expression);
repeat_until_cycle = "REPEAT" , (statement block_statements) , "UNTIL" , repeat_until_cycle_cond;	repeat_until_cycle = tokenREPEAT >> (statement block_statements) >> tokenUNTIL >> repeat_until_cycle_cond;
input = "GET" , (ident , [index_action] "(" , ident , [index_action] , ")");	input = tokenGET >> (ident >> -index_action tokenGROUPEXPRESSIONBEGIN >> ident >> -index_action >> tokenGROUPEXPRESSIONEND);
output = "PUT" , expression;	output = tokenPUT >> expression;
statement = bind_right_to_left bind_left_to_right cond_block forto_cycle while_cycle repeat_until_cycle labeled_point goto_label input output ";" ;	statement = bind_right_to_left bind_left_to_right cond_block forto_cycle while_cycle repeat_until_cycle labeled_point goto_label input output tokenSEMICOLON;

block_statements = "{" , {statement} , "}";	block_statements = tokenBEGINBLOCK >> *statement >> tokenENDBLOCK;
program = "NAME" , program_name , ";" , "BODY" , "DATA" , [declaration] , ";" , {statement} , "END";	program = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> (-declaration) >> tokenSEMICOLON >> *statement >> tokenEND;
digit = "0" "1" "2" "3" "4" "5" "6" "7" "8" "9";	digit = digit_0 digit_1 digit_2 digit_3 digit_4 digit_5 digit_6 digit_7 digit_8 digit_9;
non_zero_digit = "1" "2" "3" "4" "5" "6" "7" "8" "9";	non_zero_digit = digit_1 digit_2 digit_3 digit_4 digit_5 digit_6 digit_7 digit_8 digit_9;
unsigned_value = (non_zero_digit , {digit}) "0";	unsigned_value = ((non_zero_digit >> *digit) digit_0) >> BOUNDARIES;
value = [sign] , unsigned_value;	value = (-sign) >> unsigned_value >> BOUNDARIES;
letter_in_lower_case = "a" "b" "c" "d" "e" "f" "g" "h" "i" "j" "k" "l" "m" "n" "o" "p" "q" "r" "s" "t" "u" "v" "w" "x" "y" "z";	letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;
letter_in_upper_case = "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V" "W" "X" "Y" "Z";	letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;
ident = "_" , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case;	ident = tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;
sign = "+" "-";	sign = sign_plus sign_minus;
	sign_plus = '+' >> BOUNDARIES;
	sign_minus = '-' >> BOUNDARIES;
	digit_0 = '0';
	digit_1 = '1';
	digit_2 = '2';
	digit_3 = '3';
	digit_4 = '4';
	digit_5 = '5';
	digit_6 = '6';
	digit_7 = '7';

	digit_8 = '8';
	digit_9 = '9';
	tokenCOLON = ":" >> BOUNDARIES;
	tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;
	tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;
	tokenCOMMA = "," >> BOUNDARIES;
	tokenNOT = "NOT" >> STRICT_BOUNDARIES;
	tokenAND = "AND" >> STRICT_BOUNDARIES;
	tokenOR = "OR" >> STRICT_BOUNDARIES;
	tokenEQUAL = "==" >> BOUNDARIES;
	tokenNOTEQUAL = "!=" >> BOUNDARIES;
	tokenLESSOREQUAL = "<=" >> BOUNDARIES;
	tokenGREATEROREQUAL = ">=" >> BOUNDARIES;
	tokenPLUS = "+" >> BOUNDARIES;
	tokenMINUS = "-" >> BOUNDARIES;
	tokenMUL = "*" >> BOUNDARIES;
	tokenDIV = "DIV" >> STRICT_BOUNDARIES;
	tokenMOD = "MOD" >> STRICT_BOUNDARIES;
	tokenGROUPEXPRESSIONBEGIN = "(" >> BOUNDARIES;
	tokenGROUPEXPRESSIONEND = ")" >> BOUNDARIES;
	tokenRLBIND = ":@" >> BOUNDARIES;
	tokenLRBIND = "=: " >> BOUNDARIES;
	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;
	tokenIF = "IF" >> STRICT_BOUNDARIES;
	tokenDO = "DO" >> STRICT_BOUNDARIES;
	tokenFOR = "FOR" >> STRICT_BOUNDARIES;
	tokenTO = "TO" >> STRICT_BOUNDARIES;
	tokenDOWNTOW = "DOWNTOW" >> STRICT_BOUNDARIES;
	tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;
	tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;
	tokenBREAK = "BREAK" >> STRICT_BOUNDARIES;

	tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;
	tokenREPEAT = "REPEAT" >> STRICT_BOUNDARIES;
	tokenUNTIL = "UNTIL" >> STRICT_BOUNDARIES;
	tokenGET = "GET" >> STRICT_BOUNDARIES;
	tokenPUT = "PUT" >> STRICT_BOUNDARIES;
	tokenNAME = "NAME" >> STRICT_BOUNDARIES;
	tokenBODY = "BODY" >> STRICT_BOUNDARIES;
	tokenDATA = "DATA" >> STRICT_BOUNDARIES;
	tokenEND = "END" >> STRICT_BOUNDARIES;
	tokenBEGINBLOCK = "{" >> BOUNDARIES;
	tokenENDBLOCK = "}" >> BOUNDARIES;
	tokenLEFTSQUAREBRACKETS = "[" >> BOUNDARIES;
	tokenRIGHTSQUAREBRACKETS = "]" >> BOUNDARIES;
	tokenSEMICOLON = ";" >> BOUNDARIES;
	STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) (!(qi::alpha qi::char_("_")));
	BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY);
	BOUNDARY = BOUNDARY_SPACE BOUNDARY_TAB BOUNDARY_CARRIAGE_RETURN BOUNDARY_LINE_FEED BOUNDARY_NULL;
	BOUNDARY_SPACE = " ";
	BOUNDARY_TAB = "\t";
	BOUNDARY_CARRIAGE_RETURN = "\r";
	BOUNDARY_LINE_FEED = "\n";
	BOUNDARY_NULL = "\0";
	NO_BOUNDARY = "";
	tokenUNDERSCORE = "_";
	A = "A";
	B = "B";
	C = "C";
	D = "D";

	E = "E";
	F = "F";
	G = "G";
	H = "H";
	I = "I";
	J = "J";
	K = "K";
	L = "L";
	M = "M";
	N = "N";
	O = "O";
	P = "P";
	Q = "Q";
	R = "R";
	S = "S";
	T = "T";
	U = "U";
	V = "V";
	W = "W";
	X = "X";
	Y = "Y";
	Z = "Z";
	a = "a";
	b = "b";
	c = "c";
	d = "d";
	e = "e";
	f = "f";
	g = "g";
	h = "h";
	i = "i";

	j = "j";
	k = "k";
	l = "l";
	m = "m";
	n = "n";
	o = "o";
	p = "p";
	q = "q";
	r = "r";
	s = "s";
	t = "t";
	u = "u";
	v = "v";
	w = "w";
	x = "x";
	y = "y";
	z = "z";